

Question 1: Find the linear regression equation for the given data:

x	y
3	8
9	6
5	4
3	2

we know,

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$c = \bar{y} - m\bar{x}$$

Here, $\bar{x} = \frac{3+9+5+3}{4} = 5$

$$\bar{y} = \frac{8+6+4+2}{4} = 5$$

x	y	$y_i - \bar{y}$	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})(x_i - \bar{x})$
3	8	3	-2	4	-6
9	6	1	4	16	4
5	4	-1	0	0	0
3	2	-3	-2	4	6

$$\Sigma = 24 \quad \Sigma = 4$$

$$\therefore m = \frac{4}{24} = 0.17$$

$$c = \bar{y} - m\bar{x}$$

$$= 5 - 0.17 \times 5$$

$$= 4.15$$

so, $y = 0.17x + 4.15$ Ans

Gradient descent

x	x	y_0	y_1	y_2	
1	1	0	0.1	0.19	
2	2	0	0.175	0.75	
3	3	0	0.25	0.91	
4	4	0	0.325	1.054	

Initially assume,

$m = 0$
 $c = 0$

so $\hat{y} = mx + c$
 $= 0 \times x + 0 = 0$

we know,

$$m := m - \alpha \frac{\partial}{\partial m} J(m, c)$$

$$c := c - \alpha \frac{\partial}{\partial c} J(m, c)$$

$$\frac{\partial}{\partial m} J(m, c) = -\frac{1}{n} \sum (y_i - \hat{y}^a) x_i$$

$$\frac{\partial}{\partial c} J(m, c) = -\frac{1}{n} \sum (y_i - \hat{y})$$

so,

$$\frac{\partial}{\partial m} J(m, c) = -\frac{1}{4} (1 + 4 + 9 + 16)$$

$$= -\frac{30}{4}$$

$$= -7.5$$

$$\frac{\partial}{\partial c} J(m, c) = -\frac{1}{4} (1 + 2 + 3 + 4)$$

$$= -2.5$$

$$m := 0 - 0.01 \times (-7.5) \quad \alpha = 0.01$$

$$\therefore m = 0.075$$

$$c = 0.025$$

$$\hat{y}_i = 0.075x + 0.025$$

$$\therefore \hat{y}_1 = 0.075 + 0.025 = 0.1$$

$$\hat{y}_2 = 0.175$$

$$\hat{y}_3 = 0.25$$

$$\hat{y}_4 = 0.325$$

again,

$$\frac{\partial}{\partial m} J(m, c) = -\frac{1}{n} \sum (y_i - \hat{y}) x_i$$

$$= -\frac{1}{4} \left\{ (1-0.1) \times 1 + (2-0.175) \times 2 + (3-0.25) \times 3 + (4-0.325) \times 4 \right\}$$

$$= - \frac{0.9 + 3.65 + 8.25 + 14.7}{4}$$

$$= -6.875$$

$$\frac{\partial}{\partial c} J(m, c) = -\frac{1}{n} \sum (y_i - \hat{y}^n)$$

$$= -\frac{1}{4} \left\{ (1-0.1) + (2-0.175) + (3-0.25) + (4-0.325) \right\}$$

$$= - \frac{0.9 + 1.825 + 2.75 + 3.675}{4}$$

$$= -2.2875$$

$$m := 0.075 - 0.01(-6.875)$$

$$\begin{aligned} m &:= 0.075 + 0.06875 \\ &= 0.14375 \end{aligned}$$

$$\begin{aligned} c &:= 0.025 - 0.01(-2.2875) \\ &= 0.47875 \end{aligned}$$

$$\hat{y}_i = 0.14375x + 0.47875$$

$$\hat{y}_1 = 0.191625$$

$$\hat{y}_2 = 0.76625$$

$$\hat{y}_3 = 0.91$$

$$\hat{y}_n = 1.0535$$

Cost function.

$$J(m, c) = \frac{1}{2m} \sum_{i=1}^m (y_i - \hat{y})^2$$

Logistic

cost function.

$$- \frac{1}{n} \sum_{i=1}^n \{ y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \}$$

So, why it is log likelihood

function / maximum likelihood

function?

we know that,

$\hat{y}_i = \sigma(z)$, it is the probability

of being 1. So probability of being 0 will be $(1 - \hat{y}_i)$

So, For the given binary output data, we need to perform $\log(\hat{y}_i)$ for the 1 and $\log(1 - \hat{y}_i)$ for the 0.

In our loss function we can see, for $y = 1$ the right portion of the equation

become 0 and only remains

$$\log(\hat{y}_i)$$

and for $y=0$ the left portion

becomes 0 and remains

$$(1-0) \log(1-\hat{y}_i)$$

so, it will tell us how likely the current model is for the given data. Then we will update the parameters to maximize the likelihood of the model.

Math

In linear regression,

$$\hat{y}_i^n = mx + c$$

In logistic:

$$\hat{y}_i = \frac{1}{1 + e^{-(mx+c)}}$$

and rest are same.

$$m' = m - \alpha \frac{\partial}{\partial m} J(m, c)$$

$$c' = c - \alpha \frac{\partial}{\partial c} J(m, c)$$

$$\frac{\partial}{\partial m} J(m, c) = -\frac{1}{n} \sum (y_i - \hat{y}) x_i$$

$$\frac{\partial}{\partial c} J(m, c) = -\frac{1}{n} \sum (y_i - \hat{y})$$

x	y	$\hat{y}_0$	$\hat{y}_1$	
2	0	0.5	0.5436	
3	0	0.5	0.5635	
5	1	0.5	0.6077	
7	1	0.5	0.6485	

assume
 $m=0$
 $c=0$

$$\hat{y}_0 = \frac{1}{1 + e^{-(0+0)}} = 0.5$$

now,

$$\alpha = 0.1$$

$$m := m - \alpha \frac{\partial}{\partial m} J(m, c)$$

$$\frac{\partial}{\partial m} J(m, c) = -\frac{1}{n} \sum (y_i - \hat{y}_i^m) x_i$$

$$= -\frac{1}{4} (-0.5 \times 2 - 0.5 \times 3 + 0.5 \times 5 + 0.5 \times 7)$$

$$= -\frac{-1 - 1.5 + 2.5 + 3.5}{4}$$

$$= -\frac{3.5}{4} = -0.875$$

$$\frac{\partial}{\partial c} J(m, c) = -\frac{1}{n} \sum (y_i - \hat{y}_i^m)$$

$$= -\frac{1}{4} (-0.5 - 0.5 + 0.5 + 0.5)$$

$$= 0$$

$$\begin{aligned} \text{so, } m &:= 0 - 0.1 \times (-0.875) \\ &= 0.0875 \end{aligned}$$

$$c := 0 - 0.1 \times 0 = 0$$

$$\begin{aligned} \text{So, } \hat{y}_1 &= \frac{1}{1 + e^{-(0.0875 \times 2)}} \\ &= \frac{1}{1 + 0.830457} \\ &= 0.5436 \end{aligned}$$

$$\begin{aligned} \hat{y}_2 &= \frac{1}{1 + e^{-(0.0875 \times 3)}} \\ &= 0.5653 \end{aligned}$$

$$\hat{y}_3 = \frac{1}{1 + e^{-(0.0875 \times 5)}} = 0.6077$$

$$\hat{y}_4 = \frac{1}{1 + e^{-(0.0875 \times 7)}} = 0.6485$$

